



Anime Recommendation System

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INTRODUCTION

Our solution:

An intelligent ML-based anime recommendation system.

- recommends similar anime
- suggests anime by genre
- predicts popularity



Problem

- Anime platforms have a large number of titles
- Users struggle to find relevant anime
- Manual searching is time-consuming
- Recommendation systems solve this problem

OBJECTIVES

- This project will develop an AI-powered recommendation system that suggests anime to users based on user ratings and anime metadata.
- Use Machine Learning for popularity prediction
- Recommend anime based on:
 - Content similarity
 - Genre
 - Search keywords
- Provide an interactive web interface using Streamlit

DATASET DESCRIPTION

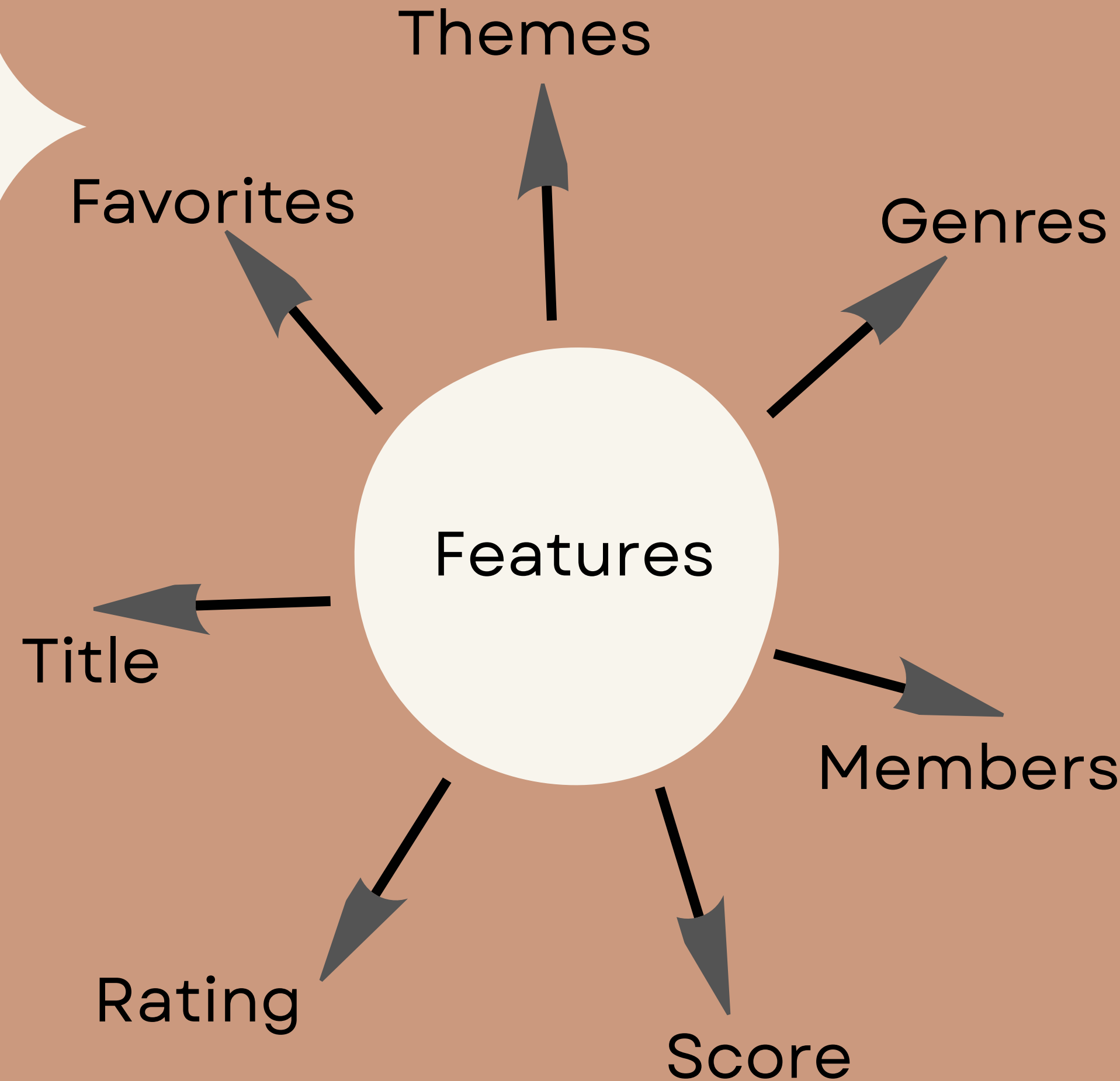
Source: MyAnimeList Dataset

Number of records: ~500 anime

Data type: Textual + Numerical

Tools Used

- Python
- Scikit-Learn
- Streamlit
- Pandas, NumPy



Data Preprocessing

- Loaded the anime dataset using Pandas
 - Selected only relevant features for the model
 - (Title, Description, Genres, Themes, Rating, Score, Members, Favorites)
 - Removed unnecessary columns to reduce noise
 - Handled missing values:
 - Replaced missing text values with empty strings
 - Converted numerical columns and replaced invalid values with 0
 - Converted data into proper numerical format for machine learning
 - Combined text features to prepare input for TF-IDF vectorization
-

MACHINE LEARNING MODELS USED

Model 1: TF-IDF Vectorizer (Unsupervised Learning)

- Converts anime text into numerical vectors
- Uses:
 - Title
 - Description
 - Genres
 - Themes
- Similarity calculated using cosine similarity
- No accuracy

Model 2: Logistic Regression (Supervised Learning)

- Predicts anime popularity
- Input features:
 - Score
 - Members
- Output:
 - Popular / Not Popular

MODEL TRAINING & EVALUATION

```
Training TF-IDF recommendation model...
TF-IDF model trained and saved
Training popularity prediction model...
```

```
Popularity model classification report:
```

	precision	recall	f1-score	support
Not Popular	0.79	0.57	0.66	782
Popular	0.64	0.83	0.72	718
accuracy			0.70	1500
macro avg	0.71	0.70	0.69	1500
weighted avg	0.72	0.70	0.69	1500

```
Overall accuracy: 0.70
Popularity model saved
```

```
MODEL PERFORMANCE SUMMARY
```

```
-----
TF-IDF Recommendation Accuracy : N/A (Unsupervised)
Popularity Prediction Accuracy : 0.70
-----
```

Training Process

- Dataset split:
 - 80% Training
 - 20% Testing
 - Popularity defined using median of favorites
 - Evaluation Metric:
 - Accuracy
 - Classification Report
-

Result:

Popularity Prediction Accuracy $\approx$ 70%

FEATURES OF THE SYSTEM

KEY FEATURES

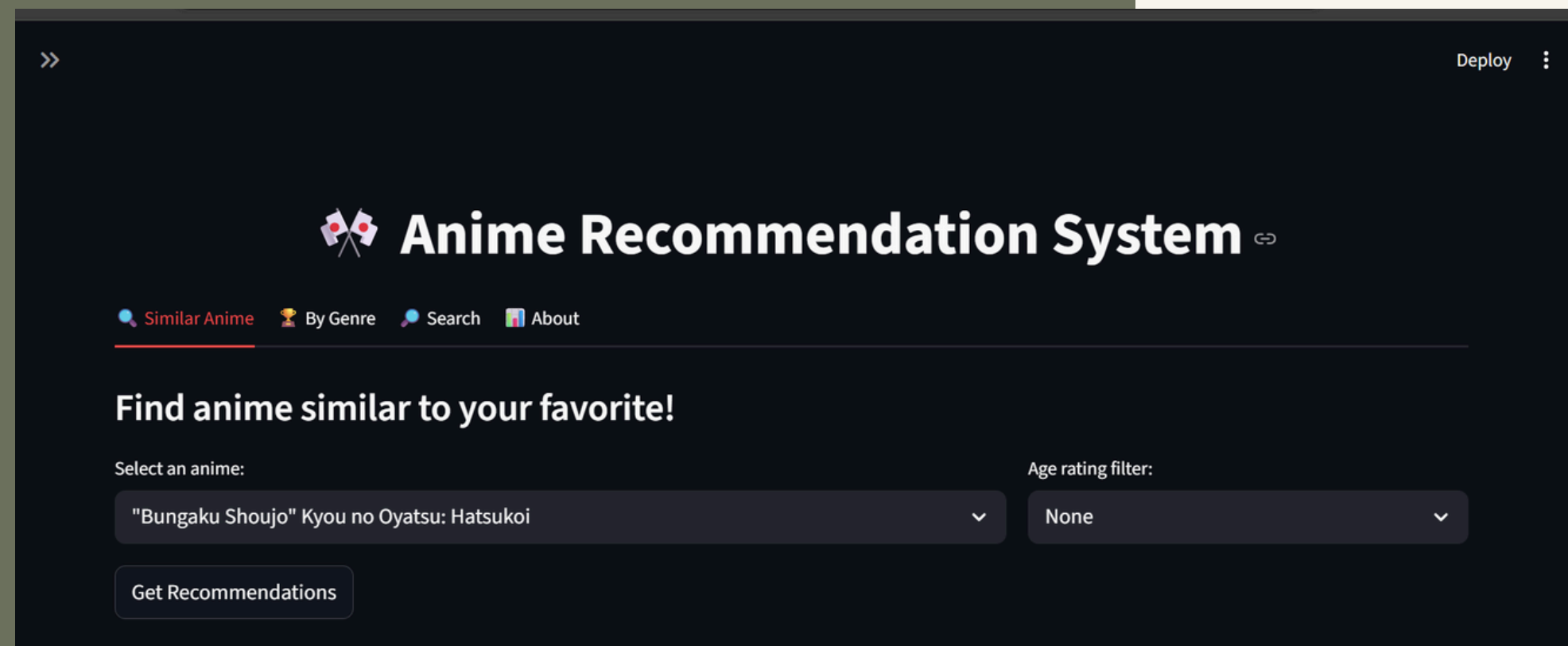
- 01 Similar Anime Recommendation
- 02 Genre-based Recommendation
- 03 Keyword Search
- 04 Age Rating Filter
- 05 Popularity Probability Prediction
- 06 Interactive Web Interface

USER INTERFACE (STREAMLIT)

Web Application

- Built using Streamlit
- User-friendly interface
- Tabs:

- Similar Anime
- Genre Recommendation
- Search
- About



RESULTS & OUTPUT

- **Sample Output**
- Displays:
- Similarity percentage
- Popularity probability
- Score and members
- Results ranked using:
- Similarity
- Popularity
- Score

>> Deploy

🌸 Anime Recommendation System

Similar Anime By Genre Search About

Find anime similar to your favorite!

Select an anime: Age rating filter:

Get Recommendations

Found 10 recommendations!

1. Akai Koudan Zillion Recaps 5.91

Genres: Action, Adventure, Sci-Fi

Type: Special | Rating: PG-13 - Teens 13 or older

Similarity	Popularity	Members
20.8%	0.1%	935

Search Genre

Found 10 anime in action!

1. Fullmetal Alchemist: Brotherhood	Score	Members
Action, Adventure, Drama, Fantasy	9.1	0K

```
Choose an option:
[1] Recommend anime similar to a given title
[2] Recommend top anime by genre
[3] Search for titles containing a keyword
[0] Exit
Enter your choice: 1
Enter anime title: a kite
Age rating filter (PG-13, R, or Enter to skip):

2636. Akai Koudan Zillion Recaps
Genres      : Action, Adventure, Sci-Fi
Type        : Special
Rating      : PG-13 - Teens 13 or older
Score       : 5.91
Favorites   : 0.0
Popularity   : 0.1%
```

```
Choose an option:
[1] Recommend anime similar to a given title
[2] Recommend top anime by genre
[3] Search for titles containing a keyword
[0] Exit
Enter your choice: 2
Enter a genre: action

4392. Fullmetal Alchemist: Brotherhood
Genres      : Action, Adventure, Drama, Fantasy
Type        : TV
Score       : 9.1
Members     : 0.0
```

Ethics & Bias

- Dataset may favor popular anime
- Older or less-known anime may get fewer recommendations
- The system does not collect personal user data
- No harmful or adult filtering is built-in
- Future improvements should include:
 - Fairness
 - User preferences
 - Content safety

FUTURE IMPROVEMENTS

- 01 Add neural networks
- 02 Hybrid system (content + user behavior)
- 03 Add user-based recommendations
- 04 Improve accuracy
- 05 Add real-time user feedback

Conclusion



This project successfully uses AI techniques like TF-IDF and Logistic Regression to build an anime recommendation system. It provides accurate, fast, and user-friendly anime suggestions using machine learning.

